

University of Cape Town
Department of Electrical Engineering
EEE3094S - Control Systems Engineering

Problem Set 4.1: Specifications in the S-Plane Solutions

Question 1

Answer the following questions about the performance of the second-order systems with the given poles.

- i) How long will it take for the system to settle to within 5% of its final value?
- ii) What is the damping condition of the system (i.e. over-, critically or underdamped?) and what is the damping ratio.
- iii) What is the damped frequency of the system? (does not apply to every damping condition)
 - a) $-2 \pm 3j$
 - b) -4 and -8
 - c) Repeated pole at -1
 - d) 0 and -1
 - e) $-2 \pm j$

Solution:

Solutions

a) $-2 \pm 3j$

i)

$$T_{2\%} \cong 4\tau \quad (1)$$

$$T_{5\%} \cong 3\tau \quad (2)$$

$$\tau = \frac{1}{\sigma} = \frac{1}{2} \quad (3)$$

$$T_{5\%} = \frac{3}{2} \text{ s} \quad (4)$$

ii)

- Overdamped – two poles on the real axis
- Critically damped – Two poles at the same point on the real axis
- Underdamped – Complex conjugate pair of poles

Underdamped

iii) Damped frequency is the complex component

$$\omega_d = 3 \text{ rad/s}$$

b) **−4 and −8**

i) The slower pole will be the more dominant response:

$$T_{5\%} = \frac{3}{4} \text{ s}$$

ii) Overdamped

iii) N/A

c) **Repeated pole at −1**

i) $T_{5\%} = 3 \text{ s}$

ii) Critically damped

iii) N/A

d) **0 and −1**

i) Pole at 0 is an integrator. Will result in a ramp response to a step

e) **$-2 \pm j$**

i) $T_{5\%} = \frac{3}{2} \text{ s}$

ii) Underdamped

iii) $\omega_d = 1 \text{ rad/s}$

Question 2

Sketch the region of the s-plane that the poles must fall into to satisfy the following specifications:

- Settling to within 2% of the final value within 5 seconds
- A damping ratio of more than 0.8
- A damping ratio of more than 0.5 and damped frequency below 10 rad/s

Solution:

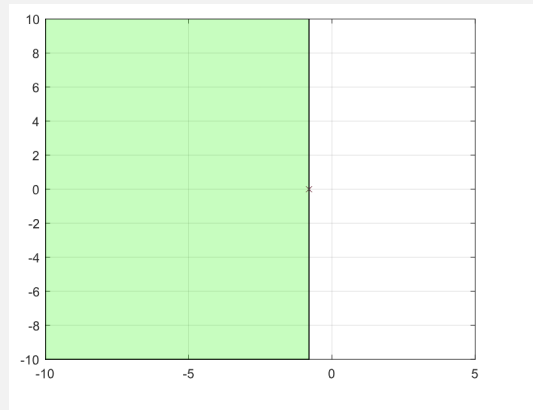
Solution

a) Settling to within 2% of the final value within 5 seconds

$$T_{2\%} = 4\tau \quad (5)$$

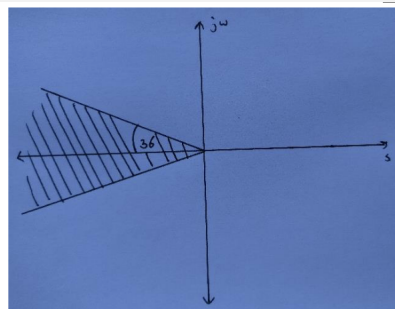
$$\tau = \frac{5}{4} \quad (6)$$

$$T = \frac{4}{5} \quad (7)$$



b) A damping ratio of more than 0.8

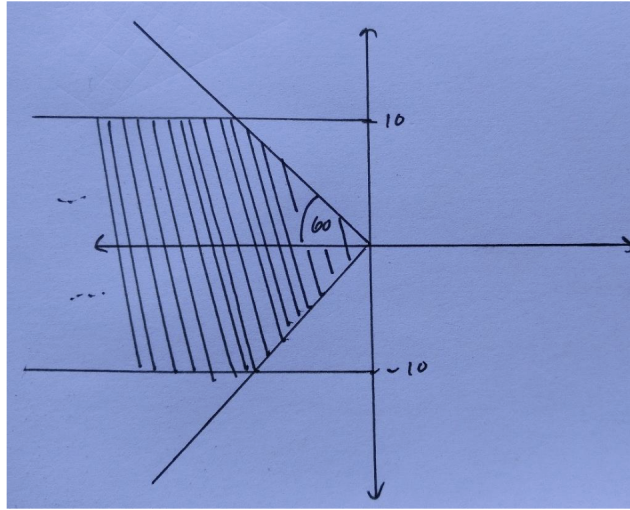
$$\theta \geq \arccos(\zeta) \geq 36.9$$



A damping ratio of more than 0.5 and 1.1 times

c) A damping ratio of more than 0.5 and damped frequency below 10 rad/s

$$\theta \geq 60$$



Question 3

A spring-mass-damper system is represented by the differential equation

$$m\ddot{x}(t) = -b\dot{x}(t) - kx(t) + f(t) \quad (8)$$

where $x(t)$ is the horizontal position of the mass and $f(t)$ is an input force. For a damping coefficient of 8 and a spring constant of 12,

- select the lightest mass that will guarantee that the system settles to 5% of its final value within 2 seconds.
- Same as (a), but if the damping coefficient has a 10% tolerance.
- Work out the damping ratio of the system for your answer in (b).
- If the system is placed in a simple feedback loop (no controller, no sensor block), work out the sensitivity function for the system in terms of m , b and k . Which parameter affects the robustness of the closed-loop system the most? Does this change for low vs. high frequencies? (hint: you can input a frequency ω to a transfer function by substituting $s = j\omega$)

Solution:

a) Select the lightest mass for settling within 2 seconds

The time constant τ of a second order system is a function of the damping ratio and natural frequency

$$\tau = \frac{1}{\zeta\omega_n} = \frac{b}{2m\omega_n} \quad (9)$$

$$\omega_n = \frac{b}{2m} \quad (10)$$

$$T_{5\%} \geq 3\tau \quad (11)$$

$$\tau \leq \frac{2}{3} \text{ s} \quad (12)$$

$$\frac{b}{2m} \leq \frac{2}{3} \quad (13)$$

$$4m \geq 24 \quad (14)$$

$$m \geq 6 \text{ kg} \quad (15)$$

b) Same as (a), but with 10% tolerance on damping coefficient

To make sure the settling time is achieved you need to take the worst case value of b

$$m \geq \frac{3b}{4}$$

Let $b = 8.8$

$$m \geq 6.6 \text{ kg}$$

c) Damping ratio for answer in (b)

$$\zeta = \frac{b}{2m\omega_n} = \frac{b}{2m} \sqrt{\frac{k}{m}} = 0.45 \quad (16)$$

d) Sensitivity function analysis

First we need the sensitivity function of the system which is ($m = 6$, $b = 8$, $k = 12$)

$$S = \frac{1}{1 + G(s)} = \frac{1}{1 + \frac{1}{ms^2 + bs + k}} \quad (17)$$

$$= \frac{ms^2 + bs + k}{ms^2 + bs + k + 1} \quad (18)$$

To see how sensitivity is affected at different frequencies we can let $s = j\omega$ and let $\omega \rightarrow 0$ and $\omega \rightarrow \infty$

$$S_{j\omega \rightarrow 0} = \frac{k}{k + 1} \quad (19)$$

$$S_{j\omega \rightarrow \infty} = 1 \quad (20)$$

Question 4

a) Component resistance sensitivity

Some component's resistance has a sensitivity of 0.02 with respect to the temperature. If its resistance is 1 k Ω at 30°C, what will its resistance be if the temperature increases by 5°?

Solution:

The fractional change in temperature is $\frac{\delta T}{T} = \frac{5}{30}$

$$S = \frac{\frac{\delta R}{R}}{\frac{\delta T}{T}} \quad (21)$$

$$\delta R = \frac{RS\delta T}{T} = 3.33 \, \Omega \quad (22)$$

$$R = 1.003 \, \text{k}\Omega \quad (23)$$

b) Sensitivity calculation from measurements

Two quantities, x and y , are measured at two instances and found to have the following values:

Quantity	Measurement 1	Measurement 2
x	10	15
y	30	70

What is the sensitivity of y with respect to x ? Would you describe y as sensitive to x ?

Solution:

$$S_x^y = \frac{\frac{\delta y}{y}}{\frac{\delta x}{x}} = \frac{\frac{40}{30}}{\frac{5}{10}} = \frac{8}{3} = 2.67$$